D2 Integral calculus: problems

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Three sections: Practice (problems 1–12), CE-level (*centraal-examenniveau*, problems 13–22) and Challenge (problems 23–28). "Exact" means: no decimals and no numerical integration on the graphing calculator (GR, *grafische rekenmachine*). Where a question gives points, such as (4p), aim for about 1.5 minutes per point, as in the central exam (CE, *centraal examen*).

Practice

Problem 1. Find an antiderivative of each function.

a. $f(x) = 6x^2 - 4x + 3$
b. $f(x) = \dfrac{1}{x^3}$
c. $f(x) = \sqrt{x}$
d. $f(x) = \dfrac{3}{x}$
e. $f(x) = e^{2x+1}$
f. $f(x) = 2^x$
g. $f(x) = \sin(3x)$
h. $f(x) = \cos\left(\frac{1}{2}x - 1\right)$

Problem 2. Use the linear substitution rule to find an antiderivative.

a. $f(x) = (2x + 5)^4$
b. $f(x) = \dfrac{1}{3x - 2}$ (for $x > \frac{2}{3}$)
c. $f(x) = e^{-x/2}$
d. $f(x) = \dfrac{1}{\sqrt{4x + 1}}$

Problem 3. Toon aan dat $F(x) = x \ln x - x$ een primitieve is van $f(x) = \ln x$. Then find the antiderivative G of $\ln x$ with $G(1) = 0$.

Problem 4. Let $f(x) = x$ on $[0, 2]$.

a. Compute the lower sum and the upper sum with $n = 4$ equal pieces.
b. Find formulas for the lower sum L_n and the upper sum U_n with n equal pieces. (Use $1 + 2 + \cdots + m = \frac{1}{2}m(m + 1)$.)
c. Show that both tend to 2, and explain why this agrees with the area of a triangle.

Problem 5. Bereken exact.

a. $\displaystyle\int_1^4 \sqrt{x}\,dx$
b. $\displaystyle\int_0^{\pi/2} \cos x\,dx$
c. $\displaystyle\int_1^e \frac{1}{x}\,dx$
d. $\displaystyle\int_0^1 e^{2x}\,dx$

Problem 6. a. Bereken exact $\displaystyle\int_0^{2\pi} \sin x\, dx$.

b. Bereken exact the area of the region between the graph of $y = \sin x$ and the x-axis on $[0, 2\pi]$.
c. Explain in one sentence why the answers to a and b differ.

Problem 7. Gegeven zijn $f(x) = x^2$ en $g(x) = x + 2$. Bereken exact de oppervlakte van het vlakdeel ingesloten door de grafieken van f en g.

Problem 8. The region under the graph of $f(x) = \sqrt{x}$ on $[0, 4]$ is rotated around the x-axis. Bereken exact de inhoud of the solid. What well-known solid is this (its shape, not its name)?

Problem 9. Use the reverse chain rule. Check each answer by differentiating.

a. $\displaystyle\int 2x\, e^{x^2}\, dx$

b. $\displaystyle\int \sin x \cos^2 x\, dx$

c. $\displaystyle\int \frac{2x}{x^2+1}\, dx$

d. $\displaystyle\int x\,(x^2+3)^4\, dx$

Problem 10. Compute the average value of

a. $f(x) = x^2$ on $[0, 3]$;
b. $g(x) = \sin x$ on $[0, \pi]$. Give the exact value.

Problem 11. A particle moves along a line with velocity $v(t) = 3t^2 - 12t + 9$ (metres per second) for $0 \le t \le 4$.

a. Compute the displacement between $t = 0$ and $t = 4$.
b. Compute the total distance travelled between $t = 0$ and $t = 4$.

Problem 12. Bereken exact de lengte van de grafiek van $f(x) = \frac{2}{3}x^{3/2}$ op $[0, 3]$.

CE-level

Problem 13. Gegeven is de functie $f(x) = x\, e^{-x}$.

a. (3p) Toon aan dat $F(x) = -(x + 1)e^{-x}$ een primitieve is van f.
b. (3p) Bereken exact de oppervlakte van het vlakdeel ingesloten door de grafiek van f, de x-as en de lijn $x = 2$.
c. (3p) W_p is the region enclosed by the graph of f, the x-axis and the line $x = p$, with $p > 0$. Show that the area of W_p is less than 1 for every p, and that it approaches 1 as $p \to \infty$. (You may use that $p\, e^{-p} \to 0$ as $p \to \infty$.)

Problem 14. Gegeven is $f(x) = 4 - x^2$. Het vlakdeel V wordt ingesloten door de grafiek van f en de x-as.

a. (3p) Bereken exact de oppervlakte van V.
b. (5p) The horizontal line $y = a$ divides V into two parts of equal area. Bereken exact a.
c. (4p) Bereken exact de inhoud van het lichaam dat ontstaat als V wentelt om de x-as.
d. (4p) Bereken exact de inhoud van het lichaam dat ontstaat als V wentelt om de y-as.

Problem 15. Gegeven is $f(x) = \ln x$.

a. (3p) Bereken exact de oppervlakte van het vlakdeel ingesloten door de grafiek van f, de x-as en de lijn $x = e$. (Use problem 3.)

b. (4p) The region bounded by the graph of f, the y-axis, the x-axis and the line $y = 1$ is rotated around the y-axis. Bereken exact de inhoud.
c. (3p) Toon aan dat $G(x) = x(\ln x)^2 - 2x\ln x + 2x$ een primitieve is van $g(x) = (\ln x)^2$.
d. (3p) The region from part a is rotated around the x-axis. Bereken exact de inhoud.

Problem 16. Gegeven zijn $f(x) = \sin 2x$ en $g(x) = \sin x$ op het domein $[0, \pi]$.

a. (4p) Bereken exact de x-coördinaten van de snijpunten van de grafieken van f en g.
b. (6p) Bereken exact de totale oppervlakte van de vlakdelen ingesloten door de grafieken van f en g.

Problem 17. a. (4p) The region under the graph of $f(x) = \cos x$ on $[0, \frac{1}{2}\pi]$ is rotated around the x-axis. Bereken exact de inhoud. (Use $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$.)

b. (4p) The region between the graphs of $y = \cos x$ and $y = \sin x$ on $[0, \frac{1}{4}\pi]$ is rotated around the x-axis. Bereken exact de inhoud.

Problem 18. Water flows into a tank. The rate of flow in litres per minute is $q(t) = 20t\,e^{-0.1t^2}$, with t in minutes and $t \geq 0$. At $t = 0$ the tank is empty.

a. (3p) Toon aan dat $Q(t) = -100\,e^{-0.1t^2}$ een primitieve is van q.
b. (3p) Calculate how many litres flow in during the first 5 minutes. Give the exact value and the value rounded to one decimal.
c. (2p) Calculate the average rate of flow in the first 5 minutes, rounded to one decimal.
d. (3p) The tank holds 120 litres. Show that it never overflows.

Problem 19. Gegeven is $f(x) = \frac{1}{2}\left(e^x + e^{-x}\right)$. (This curve is called a catenary, *kettinglijn*: the shape of a hanging chain.)

a. (3p) Toon aan dat $1 + \big(f'(x)\big)^2 = \big(f(x)\big)^2$.
b. (4p) Bereken exact de lengte van de grafiek van f op $[-1, 1]$.
c. (2p) Compare with the area under the graph on $[-1, 1]$. What do you notice?
d. (Extra) Show that for this f, the arc length from 0 to x always equals $f'(x)$.

Problem 20. a. (3p) Write down the integral for the length of the graph of $f(x) = x^2$ on $[0, 1]$, and calculate it with the GR, rounded to three decimals.

b. (2p) Without calculating: why must the answer lie between $\sqrt{2} \approx 1.414$ and 2?

Problem 21. For every $p > 0$ the function $f_p(x) = px - x^2$ is given.

a. (4p) Show that the area enclosed by the graph of f_p and the x-axis equals $\frac{1}{6}p^3$.
b. (2p) Bereken exact voor welke p deze oppervlakte 36 is.

Problem 22. a. (3p) Toon aan dat $F(x) = -\ln(\cos x)$ een primitieve is van $f(x) = \tan x$ op $\left(-\frac{1}{2}\pi, \frac{1}{2}\pi\right)$.

b. (2p) Bereken exact $\displaystyle\int_0^{\pi/4} \tan x\,dx$, and write the answer in the form $a \ln 2$.

Challenge

Problem 23. Use Riemann sums to compute $\displaystyle\int_0^1 x^3\,dx$ from the definition. You may use (or, better, prove by induction) that $1^3 + 2^3 + \cdots + n^3 = \left(\frac{1}{2}n(n+1)\right)^2$.

Problem 24. (Integration by parts, notes §11.)

a. Show that $\int x\,e^x\,dx = (x-1)e^x + c$ and compute $\int_0^1 x\,e^x\,dx$.

b. Bereken exact $\displaystyle\int_0^{\pi} x \sin x\,dx$.

c. Use integration by parts twice to find $\int x^2 e^x\,dx$, and compare with worked example 2 in the notes.

Problem 25. (Improper integrals, notes §11.)

a. Compute $\displaystyle\int_1^{\infty} \frac{1}{x^2}\,dx$.

b. Gabriel's horn: the region under $y = \frac{1}{x}$ for $x \geq 1$ is rotated around the x-axis. Show that the volume is finite and compute it.

c. The surface area of a solid of revolution is $2\pi \int f(x)\sqrt{1+f'(x)^2}\,dx$. Show that for Gabriel's horn this is infinite. (Hint: $\sqrt{1+f'^2} \geq 1$.) So you can fill the horn with paint but you cannot paint its inside. Explain why this is not a real paradox.

Problem 26. a. Prove that if f is continuous and odd ($f(-x) = -f(x)$), then $\int_{-a}^{a} f(x)\,dx = 0$; and if f is even ($f(-x) = f(x)$), then $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$. Use an antiderivative argument or the Riemann-sum picture.

b. Bereken exact $\displaystyle\int_{-1}^{1} \left(x^3\cos x + x^2\right)dx$ (no antiderivative of $x^3\cos x$ needed).

Problem 27. a. Rotate the upper half of the circle $x^2 + y^2 = r^2$ around the x-axis and derive the volume of a sphere.

b. Derive the volume of a spherical cap (*bolsegment*) of height h cut from a sphere of radius r ($0 \leq h \leq 2r$). Check your formula for $h = r$ and $h = 2r$.

Problem 28. Suppose you only know the definition $L(x) = \displaystyle\int_1^x \frac{1}{t}\,dt$ for $x > 0$ (forget everything about ln).

a. Explain why $L'(x) = \frac{1}{x}$ and $L(1) = 0$.

b. Fix $a > 0$ and let $h(x) = L(ax) - L(x)$. Show that $h'(x) = 0$, and conclude that $L(ax) = L(a) + L(x)$.

c. Compute $\displaystyle\frac{d}{dx}\int_0^{x^2} e^{t^2}\,dt$.